

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

SENTENCE AND PARAGRAPH EMBEDDINGS

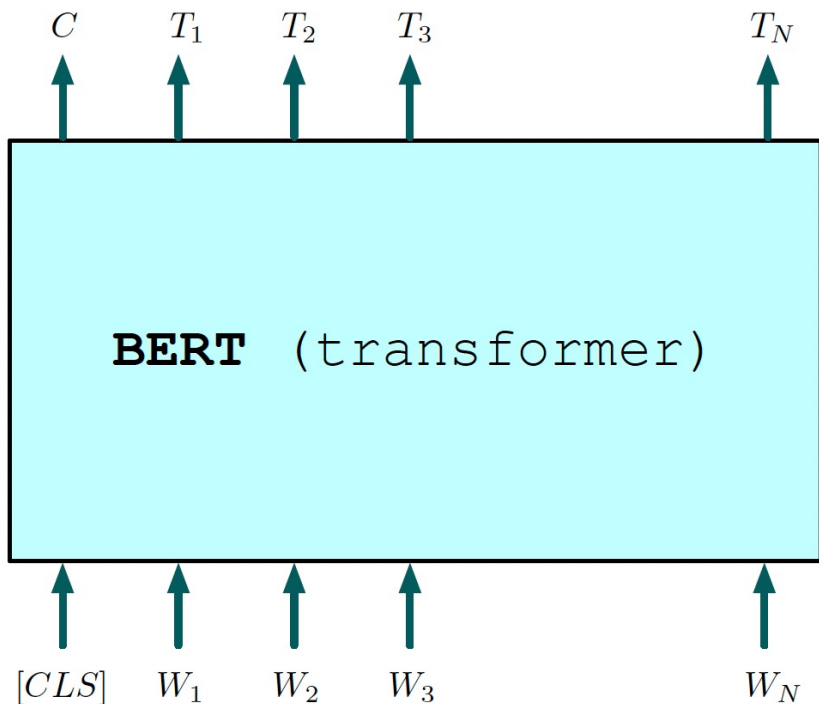
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BERT

Devlin, Chang, Lee, Toutanova (2018). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*.



- Input: Word-tokens $W_1, \dots, W_N$ (for $N \leq 512$)
- Output: Embeddings $C, T_1, \dots, T_N \in \mathbb{R}^{1024}$

Embeddings:

- C : entire sentence/paragraph
- T_j : word W_j in context

Model size: 340M parameters

Pre-training BERT

Unsupervised learning from two corpora:

- 1 English Wikipedia (2.5B words)
- 2 BooksCorpus (800M words)

Two self-supervised tasks:

- 1 Masked language model.

Figure out the missing word: *The waiter poured me a [MASK] of wine.*

- 2 Next sentence prediction.

Given two sentences: did Sentence B immediately follow Sentence A?